

SHUCHENG KANG

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EDUCATION	Harvard University, Cambridge, MA Doctor of Philosophy in Electrical Engineering 9/2023 – Now Advisor: Prof. Heng Yang Tsinghua University, Beijing, China Bachelor of Engineering in Electronic Engineering 8/2019 – 7/2023 GPA: 3.95/4.00, rank: 5/262 Hangzhou No.2 High School, Zhejiang, China High School Diploma 8/2016 – 7/2019
RELATED COURSES	Calculus (A); Linear Algebra (A); Probability Theory (A); High-Dimensional Statistics (A); Numerical Analysis (A); Convex Optimization (A); Mix-Integer Optimization (A); Semidefinite Optimization (A); General Physics (A); Introduction to Robotics (A); Deep Learning (A)
EXPERIENCE	LAAS-CNRS, Toulouse, France Research Scientist 5/2025 – 7/2025 Advisor: Prof. Jean B. Lasserre
SKILLS	C++, CUDA, Python, MATLAB
RESEARCH INTERESTS	Numerical Optimization; Numerical Linear Algebra; Convex Algebraic Geometry; Robotics I aim to provide both <i>theoretical foundations</i> and <i>computational algorithms</i> for large-scale semidefinite relaxations, enabling the next generation of decision-making systems. My current research focuses on bridging nonconvex and convex optimization in fundamental problems in robotic control and computer vision. Specifically, I: (a) establish new theoretical bounds and acceleration techniques for first-order methods in semidefinite programming through field analyses and refined perturbation analyses; (b) study special facial structures in pseudo-moment cones and Gram spectrahedra, leading to specialized algorithms for the moment-sum-of-squares (SOS) hierarchy; and (c) develop scalable SDP solvers that leverage advances in large-scale convex optimization theory, exploit problem-specific sparsity, and harness modern computing hardware such as GPUs.
RESEARCH HIGHLIGHTS	Simplicial Regularizability of the Pseudo-Moment Cone Establishes a new <i>simplicial regularizability</i> theory for the facial geometry of the <i>pseudo-moment cone</i>, showing that generically generated moment matrices with up to $\Theta(n^d)$ atoms admit simplicial minimal faces. Develops a <i>Carathéodory-type extreme-ray decomposition</i> algorithm, yielding an efficient and numerically stable atomic decomposition method for moment matrices. Submitted <i>SIAM Journal on Optimization</i>'26. [Paper] [Code]

Local Second-Order Limit Dynamics of ADMM for SDP

Builds up a local *second-order limit dynamics framework* for *slow convergence regions* in ADMM for SDP. Links key properties of the limit map to ADMM’s asymptotic behavior in these regions.

Submitted *Mathematical Programming*’26. [\[Paper\]](#) [\[Code\]](#)

Factorization-Free PSD Cone Projection via Composite Polynomial Filtering

Provides the first factorization-free PSD cone projection via a carefully designed *Composite polynomial filter inspired by homomorphic encryption*. Shows up to 10× *speed-up* compared to state-of-the-art GPU-based eigenvalue solver while achieving moderate accuracy. Package written in *C++* and *CUDA*.

Submitted *SIAM Journal on Optimization*’25 (minor revision). [\[Paper\]](#) [\[Code\]](#)

Local Linear Convergence of ADMM for SDP under Strict Complementarity

Establishes theoretical foundation for Alternating Direction Method of Multiplier (ADMM)’s *local linear convergence rate* in semidefinite programming (SDP) under strict complementarity. Offers *the first numerical observation* for ADMM’s sharp local linear convergence rate over extensive datasets.

Submitted *Mathematical Programming*’25 (major revision). [\[Paper\]](#) [\[Code\]](#)

Global Contact-Rich Planning with Sparsity-Rich Semidefinite Relaxation

Shows that contact-rich motion planning is also *sparsity-rich* when viewed as polynomial optimization (POP). Introduces the *Sparse Polynomial Optimization Toolbox (SPOT)* — implemented in C++ with interfaces to both Python and Matlab — that automates sparsity exploitation for robotics and beyond.

Accepted by *Robotics: Science and Systems (RSS)*’25. [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#)

Fast and Certifiable Trajectory Optimization

A powerful framework to compute *certifiably optimal solutions* for *nonconvex trajectory optimization* problems defined by polynomial objectives and constraints. The *first GPU-based* large-scale SDP solver implemented in GPU. Achieves up to 10× *speed-up* compared to existing solvers and runs in real time when combined with data-driven warm-starts. Package written in *C++* and *CUDA*.

Accepted by *The 16th International Workshop on the Algorithmic Foundations of Robotics (WAFR)*’24. [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#)

TALKS

Local Second-Order Limit Dynamics of ADMM for SDP

INFORMS Optimization Society Conference (IOS) 2026

Local Linear Convergence Rate of ADMM for SDP under SC

Invited Talk, Southern University of Science and Technology 2025

International Conference on Continuous Optimization (ICCOPT) 2025

POP Seminar, LAAS-CNRS 2025

Department of Mathematics Colloquium, Hong Kong Baptist University 2025

Fast and Certifiable Trajectory Optimization

International Conference on Continuous Optimization (ICCOPT) 2025

“Safe Autonomy” Workshop, Robotics: Science and Systems 2024

Lightening Talk by Early Career Participants

Oberwolfach Research Institute for Mathematics 2024

Verification and Synthesis of Robust Control Barrier Functions: Multilevel Polynomial Optimization and Semidefinite Relaxation

Conference on Decision and Control 2023

PUBLICATIONS Preprint

- [P1] Shucheng Kang and Heng Yang. Simplicial regularizability of the pseudo-moment cone and Carathéodory-type atomic decomposition of moment matrices. *SIAM Journal on Optimization*. ([pdf](#)) ([code](#))
- [P2] Shucheng Kang and Heng Yang. Local second-order limit dynamics of the alternating direction method of multipliers for semidefinite programming. Submitted to *Mathematical Programming*. ([pdf](#)) ([code](#))
- [P3] Shucheng Kang, Haoyu Han, Antoine Groudiev, and Heng Yang. Factorization-free Orthogonal Projection onto the Positive Semidefinite Cone with Composite Polynomial Filtering. Submitted to *SIAM Journal on Optimization* (minor revision). ([pdf](#)) ([code](#))
- [P4] Shucheng Kang, Xin Jiang, and Heng Yang. Local linear convergence of the alternating direction method of multipliers for semidefinite programming under strict complementarity. Submitted to *Mathematical Programming* (major revision). ([pdf](#)) ([code](#))

Journal

- [J1] Lei Huang, Shucheng Kang, Jie Wang, and Heng Yang. Sparse Polynomial Optimization with Unbounded Sets. *SIAM Journal on Optimization*. 2025 ([pdf](#)) ([code](#))
- [J2] Tianhao Wei, Shucheng Kang, Weiye Zhao, and Changliu Liu. Persistently Feasible Robust Safe Control by Safety Index Synthesis and Convex Semi-infinite Programming. *IEEE Control Systems Letters*. 2022 ([pdf](#))

Conference

- [C1] Shucheng Kang, Guorui Liu, and Heng Yang. Global contact-rich planning with sparsity-rich semidefinite relaxations. *Robotics: Science and Systems (RSS)*. 2025 ([pdf](#)) ([web](#)) ([code](#))
- [C2] Yulin Li, Haoyu Han, Shucheng Kang, Jun Ma, and Heng Yang. On the surprising robustness of sequential convex optimization for contact-implicit motion planning. *Robotics: Science and Systems (RSS)*. 2025 ([pdf](#)) ([web](#)) ([code](#))
- [C3] Shucheng Kang, Xiaoyang Xu, Jay Sarva, Ling Liang, and Heng Yang. Fast and Certifiable Trajectory Optimization. *International Workshop on the Algorithmic Foundations of Robotics (WAFR)*. 2024 ([pdf](#)) ([web](#)) ([code](#))
Best paper finalist, IEEE RAS TC on Model-based Optimization for Robotics, 2024
- [C4] Shucheng Kang, Yuxiao Chen, Heng Yang, and Marco Pavone. Verification and Synthesis of Robust Control Barrier Functions: Multilevel Polynomial Optimization and Semidefinite Relaxation. *IEEE Conference on Decision and Control (CDC)*. 2023 ([pdf](#))

- [C5] Tianhao Wei, Shucheng Kang, Ruixuan Liu, and Changliu Liu. Zero-shot Transferable and Persistently Feasible Safe Control for High Dimensional Systems by Consistent Abstraction. *IEEE Conference on Decision and Control (CDC)*. 2023 ([pdf](#))
- [C6] Xiang Zhu, Shucheng Kang, and Jianyu Chen. A Contact-safe Reinforcement learning Framework for Contact-rich Robot Manipulation. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022 ([pdf](#))

TEACHING	Teaching Fellow, (ES 155) Systems and Control, School of Engineering and Applied Sciences, Harvard University, Fall 2024
	Teaching assistant, Programming Basics, Electronic Engineering, Tsinghua University, Summer 2021
SERVICES	Reviewer, IEEE Transactions on Automatic Control (TAC)
	Reviewer, IEEE Conference on Decision and Control (CDC), 2025
	Reviewer, The International Journal of Robotics Research (IJRR)
	Reviewer, Mathematical Programming Computation (MPC)
	Reviewer, Journal of Scientific Computing
	Reviewer, Mathematical Methods of Operations Research
	Reviewer, Robotics and Automation Letters (RA-L)
	Reviewer, Transactions on Machine Learning Research (TMLR)
	Reviewer, IEEE Control Systems Letters (L-CSS)
	Reviewer, American Control Conference (ACC), 2023
HONORS AND AWARDS	Best paper finalist, IEEE RAS TC on Model-based Optimization for Robotics, 2024
	Comprehensive Excellence Award, (3%), 2022
	National Scholarship, (highest student award in Chinese universities, 2%), 2021